

Week 3: HCF (Highest Common Factor)

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

HCF is useful whenever we need to divide or share items into the largest possible equal groups — for example, arranging students into the biggest equal teams, or simplifying a fraction to its lowest terms.

Definition

The **HCF** (also called GCF) of two or more numbers is the largest number that divides exactly into all of them.

Method 1 — Listing Factors

Example 1: Find the HCF of 12 and 18.

Factors of 12: 1, 2, 3, 4, 6, 12

Factors of 18: 1, 2, 3, 6, 9, 18

Common factors: 1, 2, 3, 6

HCF = 6 (the largest common one)

Example 2: Find the HCF of 20 and 30.

Factors of 20: 1, 2, 4, 5, 10, 20

Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30

Common factors: 1, 2, 5, 10

HCF = 10

Method 2 — Prime Factorization

Example: Find the HCF of 36 and 60.

$$36 = 2^2 \times 3^2$$

$$60 = 2^2 \times 3 \times 5$$

Take the **lowest power** of common prime factors: $2^2 \times 3 = 4 \times 3 = 12$

HCF = 12

Method 3 — Division Method

Example: Find the HCF of 48 and 18.

1. Divide the bigger by the smaller: $48 \div 18 = 2$ remainder 12

2. Divide 18 by the remainder 12: $18 \div 12 = 1$ remainder 6

3. Divide 12 by 6: $12 \div 6 = 2$ remainder 0

4. When remainder = 0, the last divisor is the HCF → **HCF = 6**

Real-Life Application

A teacher has 24 pencils and 36 erasers and wants to share them into the largest possible number of identical packs with no items left over. The HCF of 24 and 36 is 12, so the teacher can make 12 packs, each with 2 pencils and 3 erasers.

Vocabulary

Word	Meaning
Factor	A number that divides exactly into another number, leaving no remainder
Common factor	A number that is a factor of two or more given numbers
Lowest terms	The simplest form of a fraction, where numerator and denominator share no common factor except 1

Evaluation Questions

- 1 Find the HCF of 24 and 32.
- 2 Find the HCF of 45 and 75 using prime factorization.
- 3 Use the division method to find the HCF of 56 and 98.
- 4 Find the HCF of 16 and 40 by listing factors.
- 5 Explain, in your own words, what "HCF" means.
- 6 A trader has 18 oranges and 27 mangoes. What is the largest number of identical baskets she can make using all the fruits with none left over?
- 7 Find the HCF of 14, 21, and 35.
- 8 Why is the HCF never bigger than the smallest of the given numbers?
- 9 Find the HCF of 9 and 16. What do you notice when two numbers share no common factors besides 1?
- 10 List all the factors of 40.